



Liability of internet intermediaries and legal challenges: a comprehensive systematic review

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Abstract

This systematic literature review examines the liability of internet intermediaries and the legal challenges associated with their role in the digital ecosystem. The study begins by outlining the significance of understanding intermediary liability in light of increasing concerns regarding online content, misinformation and user protection. The problem statement highlights the inconsistencies in legal frameworks across jurisdictions, which complicate the accountability of internet intermediaries for third-party content. To achieve this, the authors conducted an extensive search of scholarly articles from reputable databases such as Scopus and Web of Science focusing on studies published between 2023 and 2025. The study follows the PRISMA framework. The database found ($n=26$) is the final primary data that has been analyzed. The findings are divided into four (4) themes, namely: (1) legal liability and regulation, (2) emerging technologies and online harm, (3) human rights and governance in the digital age, and (4) cybersecurity and digital governance. The analysis reveals that despite the challenges, there are emerging best practices that could influence future regulatory developments. In conclusion, addressing the legal challenges faced by internet intermediaries requires a collaborative effort among lawmakers, industry players and civil society to establish definitive legal guidelines that strike a balance between accountability and safeguarding freedom of expression as well as digital innovation. This review underscores the importance of ongoing research and cross-disciplinary discussion to navigate the complexity of intermediary liability in a rapidly evolving technological landscape.

Keywords Intermediary liability · Digital governance · Online content regulation · Cybersecurity and emerging technologies · Human rights in digital ecosystem

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1 Introduction

The rapid evolution of the internet has revolutionized information access and dissemination globally, leading to internet intermediaries such as Internet Service Providers (ISPs), social media platforms and search engines. These intermediaries play a crucial role in the digital ecosystem by facilitating communication and the exchange of information. However, their role has also raised significant legal challenges, particularly concerning their liability for third-party content (Brunner 2016; Polański 2018). The legal landscape governing the internet intermediary liability is complex and varies across jurisdictions, often balancing the need to protect individual rights with the imperative to uphold freedom of expression and innovation (Baltrimas 2022; van Eecke 2011; Julià-Barceló and Koelman 2000). For instance, the European Union's Directive on Electronic Commerce provides a liability regime for online intermediaries, offering them certain protections under specific conditions. This directive aims to foster a functional internal market while safeguarding freedom of speech.

However, recent case law and legislative developments have gradually eroded these protections, compelling intermediaries to monitor user activities more closely, which paradoxically contradicts the directive's original intent (van Eecke 2011; Kuczerawy 2015). This evolving legal framework has sparked debates on the appropriate balance between intermediary liability and the protection of fundamental rights (De Gregorio 2019; Keller 2017; Rimmer 2017). In the United States, Sect. 230 of the Communications Decency Act grants broad immunity to internet intermediaries, shielding them from liability for user-generated content. This provision is pivotal in maintaining an open and free internet, as it prevents intermediaries from over-censoring content in fear of legal repercussions (Logan 2019). However, this broad immunity has faced criticism for allowing harmful content to proliferate unchecked, prompting calls for reform to address these concerns without stifling free expression (Logan 2019; Wright 2017). The legal challenges surrounding internet intermediary liability are not confined to Western jurisdictions. In China, the Tort Liability Law imposes joint liability on intermediaries who knowingly allow illegal activities on their platforms without taking action to prevent them.

This has led to significant discussions on the standards of knowledge and the extent of monitoring required from intermediaries to avoid liability (Tao 2012). Similarly, in Malaysia, recent court rulings in *Peguan Negara Malaysia v Mkini Dotcom Sdn Bhd & Anor* [2021] 2 MLJ 652 (MKini), have highlighted the complexities of intermediary liability, especially where intermediaries also act as publishers, further complicating the legal landscape (Daud et al. 2022). This comprehensive systematic review aims to explore the liability of internet intermediaries and the associated legal challenges across different jurisdictions. By examining the court case mandates for legal reasoning, the impact of existing laws and the progressive attitudes towards intermediary responsibilities, this study seeks to provide a nuanced understanding of the current state of intermediary liability and propose balanced recommendations for tackling these challenges (Baltrimas 2022; Julià-Barceló and Koelman 2000; Mac Síthigh 2020).

2 Literature review

The landscape of internet intermediaries has undergone considerable transformation due to the rapid evolution of digital platforms, which serve as essential conduits for content dissemination. Bleakley et al. (2024) highlight the challenges faced by social media platforms in moderating harmful content. They emphasize the debate surrounding self-regulation versus state regulation, indicating a need for clearer guidelines to define the roles and responsibilities of internet intermediaries. Furthermore, Giamporcaro and Kuk (2024) provide insights into the affordances of digital technologies and their implications for liability, demonstrating how regulation is impacted by practices. These insights reveal the complexities and contradictions within the existing legal frameworks that govern internet intermediaries.

The concept of “duty of care,” discussed by Huang and Chen (2024), introduces an important dimension to this discourse. They argue that imposing a non-exploitative user-generated content levy on platforms could facilitate a more equitable distribution of responsibilities between content creators and intermediaries, thereby enhancing accountability without stifling innovation. The balancing act between protecting freedom of expression and ensuring accountability for harmful content is particularly delicate in Malaysia. The Federal Constitution guarantees the freedom of speech. However, this right is not absolute and must be weighed against the potential harms associated with unregulated online content. The MKini case serves as a pivotal moment that illustrates this tension, as it challenges the notion of intermediary immunity. Recent discussions underscore the necessity for clear legal frameworks that delineate the extent of liability for intermediaries, especially in cases involving third-party content.

As identified in the works of Bleakley et al. (2024), the ongoing debate over whether technology companies are responsible for moderating content complicates these problems even further, as stakeholders call for greater accountability from intermediaries while offsetting against censorship and privacy issues. In the context of regulatory challenges, de Caria (2023) examines the intersection of online speech regulation and economic freedoms, focusing on the United States. The article discusses how large platforms hosting user-generated content blur the lines between private entity regulation and public administrative oversight. This perspective is essential in understanding the regulatory landscape governing online speech, particularly as it pertains to economic initiatives. The challenges of regulating online platforms resonate with the findings of Dhali et al. (2024), who explore the need for innovative regulatory frameworks for decentralized cryptocurrencies, emphasizing the limitations of traditional command-and-control approaches.

Both studies underline the necessity of adapting existing regulatory mechanisms to cater to the evolving digital landscape. This citation political analysis is complementing the broader governance and regulation debate, as presented by Xiao (2024) and Ibrahim and Zoppolato (2024), who advocate for a more transparent and equitable policy-making approach on technology and copyright enforcement. The relationship between emerging technologies and law has become increasingly pertinent, as noted by Ibrahim and Zoppolato (2024). Their argument for a ‘law by design’ approach advocates for integrating legal principles early in the technological development process, rather than responding reactively to advancements. This proactive stance is crucial in mitigating the risks associated with new technologies, including the ethical implications of data usage and algorithmic accountability. In parallel, Gosztonyi and Lendvai (2024) critically evaluate the legal responsibilities

of online platforms, particularly in light of the recent U.S. Supreme Court rulings. Their analysis of Sect. 230 of the Communications Decency Act sheds light on the challenges of balancing free expression with the need for content regulation in digital spaces.

The regulatory landscape surrounding digital content moderation and cybercrime is undergoing significant transformations, largely influenced by evolving technological challenges and socio-political contexts. In the European Union (EU), the recognition of security threats posed by misleading political advertising and disinformation has prompted a shift from self-regulation to more stringent co-regulation frameworks. Farrand (2024) highlights how the EU's response to hybrid threats is reshaping online platform governance, advocating for regulatory measures that empower state-like actors to assert control over digital policies previously left to market forces. This regulatory mercantilism is characterized by a focus on maintaining digital sovereignty and countering perceived vulnerabilities, ultimately leading to a substantial restructuring of governance mechanisms in the digital realm. Such a paradigm shift underscores the need for comprehensive strategies that can effectively address the challenges of misleading content while ensuring that fundamental democratic principles are preserved. As the prevalence of cybercrime continues to rise globally, law enforcement agencies are compelled to reconceptualize their policing strategies to effectively combat these new threats.

The challenges associated with content moderation, particularly regarding borderline content on platforms like TikTok, reveal the complexities of governance in less-democratic contexts. Liu (2024) investigates the socio-legal intricacies of TikTok's moderation practices in South and Southeast Asia, uncovering a landscape marked by vague rationales for content governance that often reflect the political agendas of regional states. The study reveals that both platforms and governments struggle to navigate the shifting boundaries of acceptable online speech, leading to an accountability vacuum. This dynamic illustrates the significant challenges faced by regulatory bodies in balancing the competing demands of free expression and state control. Eckert et al. (2024) explore the impact of various market entry strategies on digital service innovation (DSI) in the mobility sector. Their findings indicate that early market entry and individual branding significantly influence DSI success, highlighting the importance of strategic planning in the advanced digital economy. This insight is crucial for platforms navigating the complexities of digital service delivery, where the interplay between innovation and regulation can determine market viability.

As digital technologies continue to unfold, the regulatory challenges posed by Artificial Intelligence (AI) and online intermediaries necessitate ongoing discourse and adaptive frameworks. Orwat et al. (2024) delve into the normative challenges of regulating AI, particularly concerning the protection of fundamental rights and societal values. Their analysis highlights the ambiguities inherent in defining and operationalizing risks associated with AI technologies, emphasizing the need for democratic legitimacy in regulatory choices. The discussion surrounding product liability for AI, as examined by Buiten (2024), further illustrates the complexities of assigning responsibility in AI-related incidents. The evolving nature of digital products and services demands a re-evaluation of existing legal frameworks to ensure that they adequately address the unique challenges posed by emerging technologies. The proliferation of digital platforms and social media has significantly reshaped how individuals engage in the exercise of freedom of expression. However, this transformation has also facilitated the spread of hate speech, requiring regulatory interventions to balance freedom of expression with the protection of individuals and communities.

Psychogiopoulou (2024) explores diverse challenges faced by the European Union (EU) in addressing digital hate speech, emphasizing the necessity of comprehensive regulatory frameworks to mitigate these harms. The EU's approach involves a range of instruments and tools aimed at combating online hate while fostering democratic values. Muniz Da Conceição (2024) investigates the concept of digital constitutionalism as it relates to social media governance, arguing for the application of constitutional values within private sector landscapes. Finally, addressing the barriers to research and innovation in the context of emerging technologies, Manteghi (2024) discusses the proposed Data Act's implications for text and data mining in relation to AI tools like ChatGPT. The author critiques the limitations of existing database protections, advocating for a more research-friendly regulatory environment that encourages the responsible use of machine-generated data. This discussion aligns with the overarching theme of balancing innovation with ethical and legal considerations, emphasizing the need for adaptive regulatory measures that foster both creativity and compliance.

3 Research question

Research questions are essential in a systematic literature review (SLR), as they define the review's scope, focus and structure. They guide the selection of studies, ensure relevance and reduce bias by keeping the review exhaustive and specific to the topic. Well-crafted questions help to organize and categorize data, enabling a structured analysis and synthesis of findings. In the planning phase of an SLR, defining the Research Questions (RQs) is essential and serves as the core of the entire review methodology. This framework supports the development of focused and targeted research questions that are crucial for the systematic review's coherence and relevance (Kitchenham 2007). To structure these questions, the authors applied the PICo framework, a mnemonic style particularly suited for formulating qualitative research questions, as proposed by Lockwood et al. (2015). The PICo elements are as follows:

- (1) Population (P): Refers to the group or participants central to the study, such as demographic, patient group, or community.
- (2) Interest (I): Indicates the primary focus or phenomenon under investigation, like an experience, behavior, intervention, or issue.
- (3) Context (Co): Describes the environment or setting for the population and interest, which could be geographic, cultural, social, or other relevant backdrops.

In examining liability of internet intermediaries and legal challenges, four (4) distinct themes were identified, namely: (1) legal liability and regulation, (2) emerging technologies and online harm, (3) human rights and governance in the digital age, and (4) cybersecurity and digital governance. Following the PICo framework, themes help in shaping a focused approach to the research questions. Legal liability and regulation examine how various legal frameworks define and limit the responsibility of intermediaries over user-generated content, highlighting jurisdictional complexities faced by policymakers and regulatory bodies. Emerging technologies and online harm address the legal implications of advanced technologies, such as AI and IoT, when intermediaries adopt them in the manner that may

expose users to potential harms without adequate accountability. Finally, human rights and governance in the digital age investigates the need for a balanced approach to uphold the freedom of expression while protecting human rights in digital governance practices. Based on these themes, three research questions emerge:

- (1) How do varying legal frameworks impact the liability of internet intermediaries in managing user-generated content across different jurisdictions?
- (2) What legal challenges arise from internet intermediaries' adoption of emerging technologies in relation to online harm prevention?
- (3) How do regulatory frameworks for internet intermediaries address the balance between human rights protection and freedom of expression in the digital age?

4 Materials and methods

The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) approach, as established by Page et al. (2021), is a widely recognized standard for SLR, ensuring transparency, completeness and consistency throughout the process. By following PRISMA guidelines, researchers can enhance the accuracy and rigor of their analysis with structured guidance on identifying, screening and including studies in their review. This approach emphasizes the value of randomized studies, given their capacity to reduce bias and deliver robust evidence. In this analysis, Web of Science and Scopus databases were chosen for their extensive coverage and reliability.

PRISMA's structured process consists of four (4) primary stages known as identification, screening, eligibility and data abstraction. In the identification stage, all relevant studies are located through database searches. Meanwhile, screening involves applying predefined criteria to exclude irrelevant or low-quality studies. Next, remaining studies will undergo a thorough assessment to ensure they meet inclusion criteria during eligibility phase. Finally, data abstraction extracts and synthesizes data from the included studies, which is crucial for drawing meaningful and reliable conclusions. This rigorous methodology enables a systematic review that produces dependable results, which can enlighten and guide future research and practice.

4.1 Identification

The identification stage is a crucial first step in SLR, laying the groundwork for a thorough and structured examination of existing literature. This process involved defining keywords, selecting databases and refining search terms to comprehensively cover the research topic. For this study, terms like "internet intermediaries," "liability," and "challenges" were chosen, targeting Scopus and Web of Science (WoS) for broad and high-quality coverage. Scopus offers an interdisciplinary collection of peer-reviewed research in technology, law as well as social sciences. On the other hand, WoS provided a curated selection of high-impact scholarly works. The search yielded 703 records from Scopus and 488 from WoS (as shown in Table 1), totalling 1191 publications, demonstrating the extensive research interest in intermediary liability and regulatory issues. By adopting SLR, the identification

Table 1 The search string

Scopus	TITLE-ABS-KEY (("internet intermediary*" OR internet OR "online platform") AND (liability OR "legal responsibility" OR accountability) AND challenges) TITLE-ABS-KEY (("internet intermediary*" OR internet OR "online platform") AND (liability OR "legal responsibility" OR accountability) AND challenges) AND (LIMIT-TO (PUBYEAR, 2023) OR LIMIT-TO (PUBYEAR, 2025)) AND (LIMIT-TO (DOCTYPE, "ar")) AND (LIMIT-TO (LANGUAGE, "English")) AND (LIMIT-TO (SRCTYPE, "j")) AND (LIMIT-TO (PUBSTAGE, "final")) AND (LIMIT-TO (SUBJAREA, "SOCI")) Date of Access: June 2025
WoS	("internet intermediary*" OR internet OR "online platform") AND (liability OR "legal responsibility" OR accountability) AND challenges (Topic) and 2024 or 2023 (Publication Years) and Article (Document Types) and English (Languages) and English (Languages) and Law or Telecommunications (Web of Science Categories) Date of Access: June 2025

Table 2 The selection of criterion in searching

Criterion	Inclusion	Exclusion
Language	English	Non-english
Timeline	2023–2025	< 2023
Literature type	Journal (Article)	Conference, book, review
Publication stage	Final	In press
Subject area (s)	Social science, law, telecommunication	Besides social science, law, telecommunication

phase ensured a representative, relevant dataset, forming a solid foundation for subsequent stages of the SLR.

4.2 Screening

The screening phase in the SLR process refined the initially identified literature to ensure alignment with the current research objectives. This stage involved the exclusion of studies that did not meet specific inclusion criteria, maintaining only those of high relevance and quality (refer to Table 2). Only English language publications from 2023 to 2025 were considered, ensuring consistency and access to the latest developments in internet intermediary liability and regulatory challenges. This also reflects the availability of key legal and policy analyses in English-language journals, which facilitates access to relevant literature and supports effective comparisons across jurisdictions. Additionally, certain publication types such as conference papers, book reviews, general reviews, books and items in press were excluded as they often lack of detailed insights and rigor that required for systematic reviews. To maintain focus on the most relevant disciplines, the screening was limited to studies within social science, law and telecommunication areas, which align closely with the research topic.

Duplicates were identified and removed during this phase, with a total of five duplicates excluded from the pool of relevant records. Out of the initial 1191 publications, 1130 were excluded based on these criteria, leaving 66 studies. After such removal, the dataset was refined to a final selection of 61 studies for detailed analysis. This comprehensive screening guaranteed that the final set of studies is relevant, recent and rigorously focused, establishing a strong foundation for subsequent SLR analysis.

4.3 Eligibility

The third step of SLR process is referred to as the eligibility phase, where a total of 56 articles were prepared for thorough examination. This phase is crucial as it is aimed for relevant and high-quality studies to be included in the review. Each title and key content of the articles were meticulously assessed against the predetermined inclusion criteria to verify their alignment with the specific research objectives of the current study. During this evaluation, it became evident that 30 articles did not meet the necessary qualifications for inclusion. Several factors contributed to their exclusion. Some articles were deemed irrelevant because they fell outside the designated field of study, where their focus did not pertain to the issues of internet intermediaries, liability and associated challenges.

Additionally, certain articles contain titles that were not significant enough to warrant consideration, indicating a lack of relevance to the research question. Other exclusions stemmed from abstracts that did not adequately address the objectives of the study, suggesting that the content would not contribute meaningfully to the research focus. Furthermore, some articles were eliminated due to the absence of full-text access, which essential for reviewing the empirical evidence and findings discussed within the articles. As a result of this rigorous assessment, only 26 articles remained for further review. These articles were carefully selected based on their relevance and quality, ensuring that the subsequent analysis would be grounded in reliable and pertinent literature. This careful selection process is fundamental to the integrity of the systematic review, as it lays the foundation for drawing valid conclusions and insights from the available research (as shown in Table 3).

4.4 Data abstraction and analysis

An integrative analysis served as one of the assessment strategies in this study, aimed at examining and synthesizing diverse research designs (quantitative, qualitative and mixed methods). The primary objective was to identify relevant topics and subtopics. Data collection marked the initial step in developing the themes. As illustrated in Fig. 1, the authors systematically reviewed a compilation of 26 publications to identify assertions or material pertinent to the study's topics, investigating both the methodologies and research outcomes across the studies. Following this, the author collaborated with co-authors to construct themes grounded in the evidence from the study's context. Finally, these themes were refined to ensure consistency. To validate the issues, two experts; one specializing in criminal law and the other in international law reviewed the examination phase. This expert review process reinforced the clarity, significance and adequacy of each sub-theme, establishing domain validity. Adjustments were then made based on the author's discretion, incorporating feedback and expert's recommendations.

Table 3 Number and details of primary studies database

No.	Author(s)	Title	Year	Journal	Scopus	WoS
1	Spytska (2024)	Practice-based methods of bringing to legal liability for anonymous defamation on the Internet and in the media	2024	Social and legal studios	/	
2	Zarmsky (2024)	Is international criminal law ready to accommodate online harm? Challenges and opportunities	2024	Journal of international criminal justice	/	/
3	Kaviani Johnson (2024)	Grooming and child sexual abuse in organizational settings - an expanded role for international human rights law	2024	Journal of Human rights practice	/	
4	Baroni (2023)	Artificial Intelligence in and through the media	2023	Comunicazione politica	/	
5	Fairgrieve et al. (2024)	Product liability and online marketplaces: Comparison and reform	2024	International and comparative law quarterly	/	
6	Chaudhuri and Anand (2023)	Enable the metaverse and smart society with trustworthy and sustainable 'things'	2023	Journal of data protection and privacy	/	
7	Tarr et al. (2024)	Insurance in the digital age: Legal and regulatory challenges	2024	Journal of business law	/	
8	Safitra et al. (2023)	Counterattacking cyber threats: A framework for the future of cybersecurity	2023	Sustainability (Switzerland)	/	
9	Qiu and Dwyer (2023)	Regulating Zhibo in China: Exploring multiple levels of self-regulation and stakeholder dynamics	2023	Policy and internet	/	
10	Khan (2024)	Framing online speech governance as an algorithmic accountability issue	2024	Indiana law journal	/	
11	Reviglio and Santoni (2023)	Governing platform recommender systems in Europe: insights from China	2023	Global jurist	/	
12	Dong and Chen (2024)	Meta-regulation: An ideal alternative to the primary responsibility as the regulatory model of generative AI in China	2024	Computer law and security review	/	
13	Damurski et al. (2023)	The impact of digital media on urban discourse: The journalist perspective	2023	International journal of E-planning research	/	
14	Sheeba et al. (2024)	Human rights in the digital age: Legal challenges and opportunities	2024	Library progress international	/	
15	Gosztonyi and Lendvai (2024)	Online platforms and legal responsibility: A contemporary perspective in view of the recent U.S. developments	2024	Masaryk University journal of law and technology	/	
16	Goel and Chaudhary (2024)	Securing the digital footprints of minors: Privacy implications of AI	2024	Balkan Social science review	/	
17	Peake (2024)	Challenges of using digital evidence for war crimes prosecutions: Availability, reliability, admissibility	2024	AJIL unbound	/	
18	Barnard (2023)	The serious business of having fun: EU legal protection for those working online in the digital economy	2023	International journal of comparative labour law and industrial relations	/	

Table 3 (continued)

No.	Author(s)	Title	Year	Journal	Scopus	WoS
19	Liu (2023)	The transatlantic divide: Intermediary liability, free expression, and the limits of trade harmonization	2023	International journal of law and information technology	/	
20	Ladd et al. (2023)	Cybersecurity and data protection in the European Union, the USA, and China: Does ChatGPT really make a difference?	2023	Journal for international business and entrepreneurship development	/	
21	Vicente (2023)	Protection against Disinformation on the Internet: A Portuguese Perspective.	2023	Journal of intellectual property, information technology and E-commerce law		/
22	Park and Rohatgi (2024)	Balancing the platform responsibility paradox: A case for amplification regulation to mitigate the spread of harmful but legal content online	2024	Computer law & security review		/
23	Ossa Monge (2024)	Anonymity and the challenges of regulating online harmful conducts	2024	Revista Chilena De Derecho Y Tecnologia		/
24	Krishna et al. (2023)	A comprehensive survey on pandemic patient monitoring system: Enabling technologies, opportunities, and research challenges	2023	Wireless personal communications		/
25	Kira (2025)	Regulatory intermediaries in content moderation	2025	Internet policy review, journal on internet regulation		/
26	Kohl (2024)	Toxic recommender algorithms: immunities, liabilities and the regulated self-regulation of the Digital Services Act and the Online Safety Act	2024	Journal of media law	/	

4.5 Quality of appraisal

Following the guidelines proposed by Kitchenham (2007), after selecting primary studies, it is essential to assess the quality of the research they present and compare them quantitatively. Primary studies like original research articles, papers or documents are directly included in the systematic review after the initial selection stage and serve as the main evidence sources. They are then analyzed, quality-assessed and compared either quantitatively or qualitatively to address the research questions defined for the review. In this study, the authors adopt the quality assessment (QA) method from Abouzahra et al. (2020), which include six QA criteria for our SLR. Each criterion is scored as follows: “Yes” (Y) with a score of 1 if fully met, “Partly” (P) with a score of 0.5 if only partially met and “No” (N) with a score of 0 if not met at all.

- QA1. Is the purpose of the study clearly stated?
- QA2. Is the interest and the usefulness of the work clearly presented?
- QA3. Is the study methodology clearly established?

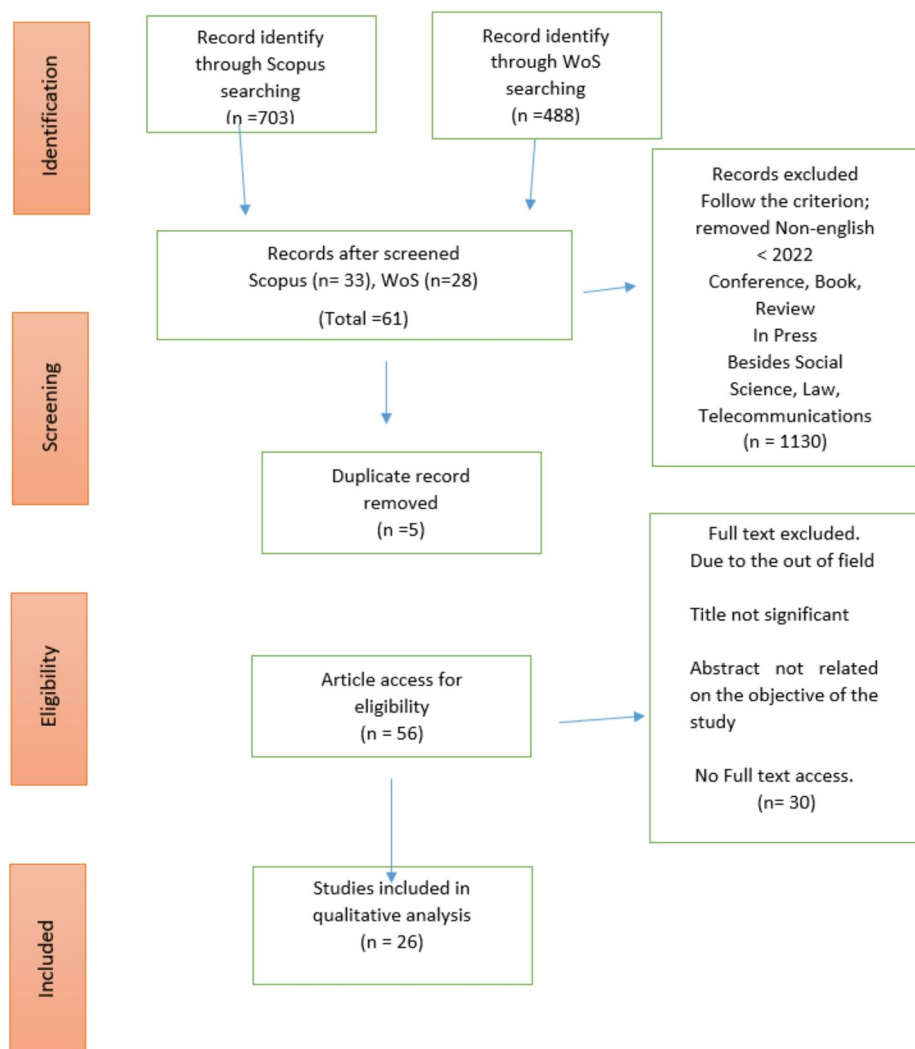


Fig. 1 Flow diagram of the proposed searching study

- QA4. Are the concepts of the approach clearly defined?
- QA5. Is the work compared and measured with other similar work?
- QA6. Are the limitations of the work clearly mentioned?

The table presents a QA process used to evaluate a study according to specific criteria. Three experts assess the study using these criteria, assigning a score of “Yes” (Y), “Partly” (P), or “No” (N) to each. This scoring threshold ensures that only studies that fulfil the required quality standard advance to the next stage. Each expert independently evaluates the study based on the given criteria and their scores are then aggregated to determine the overall rating. For a study to progress to the next stage, the combined score from all three experts must

exceed 3.0. This threshold ensures that only studies meeting a specified quality standard move forward.

1. **Is the purpose of the study clearly stated?** This criterion checks whether the study's objectives are clearly defined and articulated. A clear purpose helps to set the direction and scope of the research.
2. **Is the interest and usefulness of the work clearly presented?** This criterion evaluates whether the study's significance and potential contributions are well-explained. It measures the relevance and impact of the research.
3. **Is the study methodology clearly established?** This is to assess whether the research methodology is well-defined and appropriate for achieving the study's objectives. Clarity in methodology is crucial for the study's validity and reproducibility.
4. **Are the concepts of the approach clearly defined?** This criterion looks at whether the theoretical framework and key concepts are clearly articulated. Clear definitions are essential for understanding the study's approach.
5. **Is the work compared and measured with other similar work?** This evaluates whether the study has been benchmarked against existing research. Comparing with other studies helps position the work within the broader academic context and highlights its contributions.
6. **Are the limitations of the work clearly mentioned?** This is to ensure that the area of this study is limited within four focused themes as well as jurisdiction of selected countries by applying SLR method.

5 Result and finding

Based on QA, Table 4 shows the result of assessment performance for selected primary studies.

Based on the searching techniques, 26 articles were extracted and analyzed. All articles were categorized based on four (4) main themes, namely: (1) legal liability and regulation, (2) emerging technologies and online harm, (3) human rights and governance in the digital age, and (4) cybersecurity and digital governance.

5.1 Legal liability and regulation

The evolving landscape of internet intermediaries and their legal accountability poses significant challenges across jurisdictions, particularly as technological advancements reshape communication and information dissemination. The research by Spytka (2024) highlights the pressing need for harmonized standards regarding legal liability for anonymous defamation within online spaces. The study identifies various approaches to enforcing liability for the spread of false information, employing comparative and analytical methods to understand the complexities involved. A critical finding indicates that many national courts often dismiss claims aimed at protecting individual dignity and reputation online, resulting in a perceived immunity for disseminators of potentially defamatory content. Recommendations from this study advocate for practical measures to enhance legal accountability and

Table 4 Quality assessment (QA) for selected papers

Data	QA1	QA2	QA3	QA4	QA5	QA6	Total Mark	Percentage (%)
PS1	1	1	1	1	0.5	1	5.5	91.7
PS2	1	1	1	1	0.5	1	5.5	91.7
PS3	1	1	0.5	1	1	0.5	5	83.3
PS4	1	1	1	0.5	0.5	0.5	4.5	75
PS5	1	1	1	1	1	1	6	100
PS6	1	1	1	0.5	0.5	0.5	4.5	75
PS7	1	1	1	1	0.5	0	4.5	75
PS8	1	1	1	1	1	0	5	83.33
PS9	1	1	0.5	1	1	0.5	5	83.33
PS10	1	1	0.5	1	1	0	4.5	75
PS11	1	1	1	1	1	0.5	5.5	91.67
PS12	1	1	1	1	0.5	1	5.5	91.67
PS13	1	1	1	1	0.5	0.5	5.0	83.33
PS 14	1	1	1	0.5	0.5	1	5.0	83.33
PS15	1	1	1	1	1	0.5	5.5	91.67
PS16	1	1	1	1	0.5	1	5.5	91.67
PS17	1	1	1	0.5	0.5	0.5	5.0	83.33
PS18	1	1	1	1	1	0.5	5.5	91.67
PS19	1	1	0.5	0.5	0.5	0.5	4.5	75.00
PS20	1	1	1	1	1	1	6.0	100.
PS21	1	1	1	1	1	0.5	5.5	91.67
PS22	1	1	1	1	1	1	6	100
PS23	1	1	1	1	0.5	0.5	5	83.33
PS24	1	1	1	1	1	0.5	5.5	91.67
PS25	1	1	1	1	0.5	0.5	5	83.33
PS26	1	1	1	1	1	1	6	100

provide mechanisms for individuals to challenge anonymous defamation more effectively, thus underscoring the urgency of legislative reform in this area.

Fairgrieve et al. (2024) analyse the challenges posed by e-commerce to traditional product liability doctrines, emphasizing the need for adaptive legal frameworks. By examining various jurisdictions, the study reveals that online platforms often encounter difficulties in assuming liability for defective products sold through their sites. The authors argue that existing legal standards must evolve to encompass the unique characteristics of online commerce, proposing reforms aimed at enhancing the accountability of these marketplaces. This exploration illustrates a broader trend where legal systems struggle to keep pace with the rapid growth of digital commerce, necessitating comprehensive regulatory adjustments to safeguard consumer rights effectively. Gosztonyi and Lendvai (2024) critically examine the U.S. Supreme Court's rulings on this provision, indicating a shift in determining the boundaries of legal immunity for internet platforms. Their findings suggest that while Sect. 230 has been pivotal in fostering internet innovation, it simultaneously poses challenges for content regulation, particularly on algorithmic responsibilities related to harmful content dissemination.

This tension between free expression and the need for content control is further exacerbated by the contrasting approaches of the United States and European Union regarding intermediary responsibilities. The authors call for legislative updates that strike a bal-

ance between protecting free speech and addressing the potential harms associated with unchecked online content dissemination. The comparative analysis of intermediary liability across jurisdictions reveals stark differences in how various legal systems handle the balance between free expression and accountability. Liu (2023) explores these differences by contrasting U.S. provisions resembling Sect. 230 with the United Kingdom's more nuanced regulatory framework. This analysis underscores the unique constitutional foundations influencing each jurisdiction's approach, ultimately advocating for a diversified governance model that respects the specific legal and cultural contexts of each region.

Liu emphasizes the importance of interdisciplinary dialogue among legal and trade experts to create effective, context-sensitive regulations that uphold both free speech and essential rights such as privacy and reputation. Sect. 230 of the United States Communications Decency Act and the EU's Digital Services Act (DSA) are two variant models of intermediary liability, shaping global digital governance through the effects they have on how platforms moderate content and engage with fulfilling legal obligations. Section 230 grants broad immunity to online intermediaries for third-party content, stimulating innovation but also raising concerns on accountability. In contrast, the DSA imposes stricter care obligations, transparency, and responsiveness to illegal content, setting regulatory precedent that is the opposite of *laissez-faire* in the U.S. model. Together, these regimes generate legal asymmetries that complicate cross-border digital activity and necessitate global platforms to deal with increasingly disparate compliance environments.

On the other hand, internet intermediaries in the EU must comply with both national laws and overarching regulations like the DSA which can lead to inconsistencies in decision-making and enforcement across member states (Kira 2025). In comparing with other jurisdiction like Brazil, Kira highlights that internet intermediaries face challenges due to limited financial and human resources for effective content moderation and compliance with proposed regulations, such as Bill 2630/2020. Regulatory internet intermediaries in Brazil face legal fragility, as their structure and functions can be altered by presidential decrees, raising concerns about their independence and long-term stability. Under the DSA, decisions made by out of court dispute settlement bodies are not binding, making it limited in terms of effectiveness in resolving disputes as well as enforcing platform accountability. ODS mechanisms designed for consensual solutions, may not be well-suited for adjudicating complex rights-related disputes, especially those affecting freedom of expression.

5.2 Emerging technologies and online harm

The discourse surrounding the liability of internet intermediaries is increasingly informed by the evolving dynamics of emerging technologies and the corresponding legal challenges that accompany them. A key theme emerging from recent literature is the role of AI in transforming not only the media landscape but also the broader regulatory environment. Baroni (2023) emphasizes the dual impact of AI technologies on news media, which poses significant challenges to individual freedoms and the media's function in democratic societies. The uneven deployment of AI tools across European news media necessitates a thorough examination of regulatory proposals that aim to govern these technologies. This examination is particularly pertinent given the complexities introduced by AI integration into media operations and the associated implications for media accountability and transparency. The need for empirical investigations into the interplay between AI and media

policies becomes evident, as the regulatory frameworks must evolve to address these novel challenges effectively.

Moreover, the legal landscape is further complicated by the rapid advancement of digital technologies, which also encompasses innovations like autonomous systems and blockchain. The work of Tarr et al. (2024) highlights the implications of these technologies for the insurance industry, where traditional risk assessment models are increasingly inadequate in the face of novel cyber risks. The article discusses how AI ability to process vast datasets complicates the legal discourse around data integrity and predictive modeling assumptions. Emerging technologies are complicating intermediary liability by blurring traditional roles and responsibilities. AI-driven content moderation raises legal questions about responsibility for automated decision-making and potential bias, while the decentralization of blockchain frustrates enforcement by scattering control across anonymous nodes, undermining traditional notions of intermediary control. Autonomous systems also strain liability regimes by making decisions independently of humans, creating gaps in existing law. Ethically, these technologies demand a re-examination of due process, transparency, and fairness, specifically challenging legislators to redefine liability standards to assure accountability without stifling innovation.

The governance of platform recommender systems (PRSs) is another crucial aspect in the dialogue on internet intermediaries. Reviglio and Santoni (2023) compare regulatory efforts in Europe and China regarding PRSs, identifying significant challenges and opportunities that arise from these frameworks. The authors note that the European DSA represents a critical step toward regulating algorithmic accountability and user self-determination. However, they also point out that governance efforts must consider the diverse media environments and the complex relationship between algorithmic design and content visibility. This intersection of technology, law and social dynamics underscores the need for interdisciplinary approaches to ensure that governance frameworks are both effective and responsive to the evolving digital landscape. As the legal implications of emerging technologies continue to unfold, specific demographic groups, particularly children, require focused attention.

Goel and Chaudhary (2024) explore the privacy challenges faced by minors in an AI-driven world, where the rapid accumulation and processing of personal data pose risks to their safety. The authors advocate for comprehensive legislative measures to protect children's privacy rights, which have been inadequately addressed in existing legal frameworks. Given that children often lack awareness of the risks associated with their digital footprints, this gap in regulation calls for urgent intervention to safeguard their fundamental rights while enabling them to benefit from technological advancements. In light of the complexities surrounding harmful but legal content (HBLC) on the internet, Park and Rohatgi (2024) introduce the concept of amplification regulation, proposing a shift in the liability paradigm that holds platforms accountable for the dissemination of HBLC. This regulatory approach recognizes the unique role of recommender systems in promoting content and suggests a need for tailored strategies to balance platform responsibilities and user freedoms.

The ongoing dialogue highlights the necessity of developing legal frameworks that address the nuanced distinctions between legal and illegal content while ensuring accountability in the digital realm. Chaudhuri and Anand (2023) contribute to this theme by examining how the Internet of Things (IoT), the metaverse, and smart services are transforming digital environments. Their paper explores how new technologies can support sustainability goals and digital innovation, but also identifies critical concerns around security, privacy,

and ethical design. They argue that trust in IoT-enabled systems depends not only on technical reliability but also on regulatory structures that address user safety and data integrity. The authors stress the need for ethically aligned design and structured policy frameworks to ensure trustworthy digital governance, especially in future-facing domains like smart cities and digital twins. This analysis aligns closely with the broader challenges of emerging technologies, where technological advancement must be matched by effective safeguards to protect users and communities from evolving online risks.

The relevance of emerging technologies to public health governance is illustrated in the study by Krishna et al. (2023), which examines the use of IoT, blockchain, and machine learning in monitoring pandemic patients. The paper highlights how these technologies can aid in remote health monitoring with minimal human intervention during health crises like Covid-19. It categorizes the technological approaches into three main areas: IoT-based patient monitoring, blockchain for secure health data management, and machine learning for diagnostic analysis. While the study does not focus on intermediary liability directly, it underscores the broader implications of technological deployment for privacy, data governance, and health accountability themes increasingly relevant to intermediary responsibilities in digital health platforms. The open research questions identified by the authors also emphasize the need for legal foresight when adopting such technologies in sensitive areas like healthcare.

5.3 Human rights and governance in the digital age

The analysis of this theme reveals a multifaceted landscape shaped by the interplay of technology, human rights and legal frameworks across various jurisdictions. The issues surrounding online harm, algorithmic accountability, child protection, disinformation and data privacy illustrate the urgent need for adaptive regulatory measures that safeguard human rights while promoting technological advancement. Zarmsky (2024) highlights that these issues not only challenge the current frameworks of international criminal law (ICL) but also present opportunities for reform. The article emphasizes that ICL must evolve to address the complexities introduced by online platforms. By recognizing online harms as potential aggravating factors or new crimes within the ICL framework, a more comprehensive approach can be developed that aligns with contemporary realities.

Similarly, Kaviani Johnson (2024) addresses the issue of grooming in organizational settings, arguing for the enhancement of international human rights law to effectively combat such behaviors. The necessity of expanding legal protections for vulnerable populations, particularly children, underscores the urgency of revisiting existing frameworks to ensure robust safeguarding mechanisms in the digital age. Moreover, the governance of online speech and content moderation has emerged as a pressing concern, particularly in the context of algorithmic accountability. Khan (2024) argues that automated tools used for online speech governance are often flawed, leading to significant errors that can exacerbate the challenges of content regulation. This necessitates a shift in regulatory focus from reactive measures to a more proactive approach that emphasizes the development and deployment of these tools.

A framework grounded in algorithmic accountability could help address the complexities posed by AI technologies in both content moderation and generation. Dong and Chen (2024) further emphasize this point by advocating for a Meta-regulation model in the gover-

nance of generative AI. By incorporating diverse stakeholders, including users and dissemination platforms, this approach aims to create a more accountable regulatory environment that enhances safety and transparency while fostering innovation. The proliferation of disinformation poses another significant challenge to human rights and governance in the digital landscape. Vicente (2023) explores the complexities of managing disinformation within democratic societies, emphasizing the need for a delicate balance between protecting free speech and combating harmful misinformation.

This theme is echoed in the work of Ladd et al. (2023), who examine the regulatory measures taken by the European Union, the United States and China in response to evolving cybersecurity and data protection issues. The authors highlight the critical role that cybersecurity regulations play in safeguarding user rights while also addressing the vulnerabilities introduced by new technologies. This underscores the necessity of creating legal frameworks that not only respond to the immediate challenges of disinformation but also consider the broader implications for democratic governance. A relevant addition to this discourse is presented by Ossa Monge (2024), who examines the tension between online anonymity and the regulation of harmful online conduct. The paper highlights that while anonymity can be a tool for protecting privacy and enabling free expression, it also facilitates harmful behavior by reducing user accountability.

This duality raises critical governance challenges, especially when addressing harmful but legally ambiguous content. The authors links the debate to core human rights principles, including freedom of speech, and applies Lessig's regulatory, modalities, law, norms, and code to explore how various systems might address harmful conduct without eroding fundamental freedoms. The paper calls for a recalibration of governance strategies that preserve anonymity for legitimate use while developing safeguards against its misuse. This perspective enriches the broader human rights discussion by acknowledging the complex trade-offs in digital governance. Sheeba et al. (2024) further enrich the theme by examining how digital technologies intersect with the protection and promotion of fundamental rights. The paper identifies key challenges arising from the increasing integration of technology into daily life, especially concerning data privacy, freedom of expression, and access to information.

Through a comparative analysis of legal frameworks such as the EU's General Data Protection Regulation (GDPR) and California's Consumer Privacy Act (CCPA), the study assesses their capacity to shield users' rights against technological misuse. It also critically discusses the implications of government censorship and platform-based content regulation on online speech. On a more optimistic note, the authors highlight how technologies, particularly blockchain and social media could empower advocacy, facilitate secure communication, and promote civic engagement. Ultimately, the paper argues for a multi-stakeholder governance approach that brings together governments, industry actors, and civil society to ensure technology enhances rather than undermines human rights.

Further developing this theme, Kohl (2024) critically examines how recommender algorithms contribute to the amplification of toxic and harmful content online, and how legal systems are responding through structured regulatory interventions. The article focuses on the evolving liability regimes under the European Union's Digital Services Act (DSA) and the United Kingdom's Online Safety Act (OSA), both of which aim to shift intermediary responsibility toward risk-based content moderation. Kohl introduces the concept of "regulated self-regulation," whereby platforms are not only granted conditional immunity

but are also legally obligated to implement mechanisms for algorithmic accountability. The paper raises critical human rights concerns, particularly the risk of over-censorship and the chilling effect on freedom of expression, when platforms err on the side of caution. Through a nuanced analysis of the DSA and OSA frameworks, Kohl advocates for transparency, proportionality, and oversight in algorithmic governance. This contribution reinforces the importance of embedding human rights safeguards within digital regulation, particularly where automated content curation significantly shapes public discourse.

5.4 Cybersecurity and digital governance

The evolving landscape of cybersecurity and digital governance presents a multitude of challenges and opportunities, particularly concerning the legal responsibilities of internet intermediaries. Recent studies shed light on the implications of emerging technologies, regulatory frameworks and the impact of digital media on governance. One key area of focus is the need for robust frameworks to counteract cyber threats. Safitra et al. (2023) propose a comprehensive framework designed to enhance organizational resilience against cyber incidents, emphasizing the integration of cybersecurity capabilities with organizational leadership and innovation. This approach recognizes that as digital threats evolve, organizations must not only react but also anticipate and mitigate potential risks. In line with this, Barnard (2023) highlights the vulnerability of online workers, such as influencers and content creators, who face significant risks if their access to platforms is disrupted. The interplay between employment status and digital platform governance underscores the necessity of establishing legal protections that account for the unique challenges faced by these individuals in the digital economy.

Another critical aspect discussed in the literature pertains to the regulation of live streaming platforms, as explored by Qiu and Dwyer (2023). The study reveals the dynamic nature of regulatory practices in China, where multiple levels of self-regulation have emerged amid minimal state oversight. This decentralized regulatory landscape presents challenges in ensuring accountability and protecting users from harmful content. Moreover, the rise of digital media has raised concerns regarding the quality of information disseminated to the public, as observed by Damurski et al. (2023). Their research illustrates how local journalism is increasingly pivotal in shaping urban discourse, yet the influx of digital content complicates the task of maintaining reliability and political accountability. The legal challenges associated with digital evidence also feature prominently in recent analysis. Peake (2024) identifies several barriers to the effective use of digital evidence in war crimes prosecutions, including issues of availability, reliability and admissibility. As the landscape of digital content evolves, ensuring the integrity of evidence becomes paramount for upholding accountability in ICL.

Zarmsky (2024) discusses the potential of new technologies to both aid in the perpetration of international crimes and facilitate accountability measures. This duality emphasizes the urgent need for legal frameworks to adapt to the changing nature of online harms. Emerging technologies may increase the reliability and admissibility of digital evidence through which online material has a particularly traumatic and psychological impact on those close to the victim once it is being viewed. For example, Zarmsky (2024) discusses that the harm experienced immediately after viewing footage of torture by a victim's family member who came across the content, which then being a contributing factor to serious

Jurisdiction	Legal Instruments	Liability Framework	Regulatory Focus Areas	Recent Developments
Malaysia	Communications and Multimedia Act 1998, Section 114A Evidence Act	Presumption-based liability, rebuttable	Media responsibility, cyber offences	Post-Malaysiakini on third-party liability; lack of safe harbor
United States	Section 230 Communications Decency Act	Broad immunity	Free speech, limited state control	Reform discussions on AI content moderation and misinformation
European Union	Digital Services Act (DSA)	Conditional liability with due diligence	Systemic risk, algorithmic transparency	Enforced obligations on VLOPs, new risk frameworks
China	Cybersecurity Law, Personal Info Protection Law	Strict state-imposed liability	Content censorship, algorithmic control	Algorithm regulation, state control over AI outputs
Germany	Network Enforcement Act (NetzDG)	Conditional with take-down duties	Hate speech, timely removal	Precedent for EU-wide platform regulation
United Kingdom	Online Safety Act 2023	Duty of care model	Online harms, children's safety, illegal content	Full enforcement planned with Ofcom oversight
Australia	Online Safety Act 2021	Graduated liability framework	Cyberbullying, image abuse, youth protection	Expansion of takedown powers, online content classification
Singapore	Protection from Online Falsehoods and Manipulation Act (POFMA)	Government-directed liability	False information, national harmony	Controversial implementation via executive correction orders

Fig. 2 Comparative matrix of jurisdictional approaches to intermediary liability

mental distress, could be corroborative evidence. This is in line with ICL jurisprudence which has recognized the ‘serious mental harm’ that can be caused to third parties, particularly family or friends of the victim, when witnessing violent acts committed against another person because the Hague Court recognized that “mere sight of another person being tortured is sufficient to constitute torture in itself”.

5.5 Comparative jurisdictional approaches to intermediary liability

In light of the jurisdictional analysis presented across the thematic findings, the following table provides a clear summary of key regulatory approaches, legal instruments and emerging trends on intermediary liability in selected countries. This comparative overview improves the clarity of country’s insights and highlights the legal diversity and evolving policy responses that continue to shape digital governance globally (refer to Fig. 2).

6 Discussion and conclusion

The evolving role of internet intermediaries presents complex regulatory challenges as platforms increasingly shape communication and commerce. Significant gaps remain, particularly in accountability for anonymous defamation, where individuals have limited options to contest harmful content. Legislative reform is needed to clarify intermediary liability, as seen in global debates spurred by U.S. amendments to Sect. 230 of the Communications Decency Act. Diverging regulatory models in the U.S. and EU illustrate the need for frameworks that balance free expression with responsible content management. For Malaysia, these developments highlight the necessity of refining its own legal framework to address intermediary liability, especially in managing content and protecting user rights.

Emerging technologies, particularly AI, further complicate intermediary liability and media accountability. Using AI into news media raises concerns around transparency, fairness and its influence on public opinion, calling for further research to understand its impacts. Innovations like blockchain and autonomous systems demand regulatory reforms to mitigate cyber risks, with growing emphasis on human rights, particularly for vulnerable groups such as children. Relying more on algorithms to manage online speech brings a new set of problems. When these systems are flawed or biased, they can make unfair decisions, highlighting the need for clear rules that spell out who is responsible. As cybersecurity and digital governance keep evolve, holding platforms' accountability and protecting users is becoming even more critical. Strong leadership within organisations is key to managing these risks effectively and building public trust.

The findings of this review highlight the need for regulatory reform to address the evolving challenges of internet intermediary liability. Legal ambiguities surrounding platform immunity, anonymous defamation and third-party content continue to create uncertainty for both regulators and platforms, hence the need for clearer frameworks that protect free expression. The increasing reliance on artificial intelligence in content moderation adds further complexity, particularly where decisions are made without human supervisions raising concerns about transparency, fairness and accountability. Regulatory models must therefore clearly define the responsibilities of intermediaries using such technologies while ensuring due process and access to remedies. Beyond technical concerns, the review also points to critical human rights issues, including censorship, disinformation, data privacy and the protection of vulnerable groups. Rising cybersecurity threats and the fragility of digital evidence further reinforce the need for proactive and trustworthy governance systems. When combined, these results demonstrate the need for inclusive, risk-based and proactive policy approaches that strike a balance between innovation, accountability and fundamental rights in the digital age. These approaches were created through interaction between legislators, industry players, technical experts and the general public.

In terms of future research, a number of significant avenues become apparent. First, additional comparative research is required to examine how various legal frameworks handle internet intermediaries' responsibility, especially in jurisdictions that remain unrepresented in the body of existing literature. Second, more research is needed to evaluate the legal and ethical implications of automated procedures, particularly when they cause harm or unjust outcomes, as AI becomes more integrated into platform decision-making and content control. Third, further research is needed to determine how self-regulation and co-regulation models affect user rights over the long run, especially in environments with little govern-

mental monitoring. Future research can also look at the connections between intermediary accountability and more general concerns including protection for vulnerable users, access to remedies, and digital harm. These areas reflect key gaps identified in this review and offer valuable opportunities for strengthening legal and policy responses in the digital age.

In conclusion, the rapid advancement of the digital landscape requires a balanced regulatory approach that safeguards freedom of speech, accountability, safety and human rights. Legal frameworks must adapt to the challenges posed by emerging technologies, from intermediary liability to AI-driven risks and data privacy. A cross-jurisdictional and transparent approach can foster innovation while protecting user rights as well as building public trust in a fair and secure digital environment.

Declarations

Conflict of interest The authors declare that they have no conflicts of interest to report regarding the present study.

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